Ofqual QAN	**600/0406/X / 600/0406/X**
Quartz ID	**1053 / 1157**
Qualification	**Level 4 Diploma in Computing / Level 4 Diploma in Computing with Business Management**

Assessment	**Object-Oriented System Analysis and Design**
Assessment Type	**Global Examination**
Date	**Thursday 5th June 2025**
Time	**09.00am – 12.00pm**

Marking Scheme

This marking scheme has been prepared as a **guide only** to markers. This is not a set of model answers, or the exclusive answers to the questions, and there will frequently be alternative responses which will provide a valid answer. Markers are advised that, unless a question specifies that an answer be provided in a particular form, then an answer that is correct (factually or in practical terms) **must** be given the available marks.

If there is doubt as to the correctness of an answer, the relevant NCC Education materials should be the first authority.

Throughout the marking, please credit any valid alternative point.

Answer ALL questions

Marks

Question 1

a) Define the following terms in the context of Object-Oriented Analysis and Design and provide an example:

i) Class attributes **2**

Mark scheme
Class attributes are characteristics in the form of data variables that describe the class (1 mark) e.g. a table might be described by type of table, material from which it is made, number of legs, type of legs, height, width. (1 mark)

ii) Method **2**

Mark scheme
A method is a function that defines a behaviour of a class (1 mark) e.g. a living thing might breathe (1 mark)

iii) Abstract class **2**

Mark scheme
An abstract class is a pseudo class that cannot itself be instantiated as the characteristics of its candidate instances vary. It describes a collection of subclasses that inherit its attributes and behaviours and add some of their own specific ones (1 mark) e.g. an instance of a living thing would be a plant or animal but cannot be instantiated as these have their own additional attributes and methods. (1 mark)

b) Please select ONE (1) from the options provided for each statement below.

i) The bundling of data attributes and methods together within a single class to ensure the internal state of an object is kept private is called: **1**

A. Encapsulation
B. Refactoring
C. Abstraction
D. Inheritance

Mark scheme
A. Encapsulation (1 mark)

ii) In the SOLID design principles, the requirement for objects of a superclass to be replaceable by objects of its subclasses without affecting the correctness of a program is called: **1**

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A. Single Responsibility Principle

B. Execution of Individual Methods Principle

C. Liskov Substitution Principle

D. Dependency Inversion Principle

Mark scheme
C. Liskov Substitution Principle (1 mark)

iii) In basis path testing, a node that contains a condition is called: **1**

A. Edge

B. Predicate node

C. Region

D. Independent path

Mark scheme
B. Predicate node (1 mark)

iv) In object-oriented design, when a component has a single, well-defined responsibility, this is referred to as: **1**

A. Cohesion

B. Coupling

C. Inheritance

D. Prioritisation

Mark scheme
A. Cohesion (1 mark)

Total 10 marks

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Marks

Question 2

Scrum and Kanban are both Agile methodologies used for managing projects, but they differ in several key aspects. Describe FIVE (5) ways in which SCRUM differs from Kanban. **10**

Mark scheme
Any five from below or alternative valid answers.

	SCRUM	**Kanban**
Goal	Project management framework that uses visual tasks to workflows limiting work in progress and maximising efficiency.	Project management framework that helps teams to structure and manage their work through a set of values, principles and practices.
Focus	Focuses on time reduction.	Team commits to each increment and creates learning loops to quickly gather and join up user feedback.
Work cycle	Iterations managed by Sprints that follow Plan-Do-Check-Act (PCDA) philosophy.	Continuous flow. As soon as one task finishes next task begins.
Review	Every sprint is an opportunity to inspect and adapt. SCRUM tracks requirements and work in progress separately.	No specific mechanism to inspect and adapt. Kanban is good for work that is seen as one time effort with no need to inspect and adapt. Tends to only track implementation
Artifacts	Product backlog, sprint backlog, increment/potentially shippable product.	Kanban board, tasks. Can also include backlog with tasks not yet prioritised.
Planning	Formalised by daily SCRUM and Sprint. Start and due dates are determined by Sprint.	Teams adopt approach that suits them and the task. New task starts when a task is completed. Due dates are determined by business need.
Responsibility	Clear responsibilities assigned to Product owner, SCRUM master and developers.	Assumes groups of individuals working on tasks.
Stakeholders	Active stakeholder and customer involvement.	No mechanism provided but consultation does take place on a need to basis.
Delegation and prioritisation	Uses a 'pull system' to grab the next batch of items for each iteration. Delays / problems with any of the group of selected tasks can impact a Sprint. Best for projects with stable priorities.	Also uses a 'pull system' in the form of a systematic workflow that only allows tasks to be grabbed when the previous task has completed. Tasks are selected one at a time so they do not impact on each other. Good for projects with varying priorities.
Productivity measurement	Productivity measured by velocity through Sprints. Sprints are run back-to-back so that success of each Sprint depends on previous Sprint.	Productivity measured by 'cycle time' i.e. amount of time to complete a piece of a project from start to end.

Award 2 marks for each comparison as 1 mark for SCRUM explanation and 1 mark for Kanban.

Total 10 marks

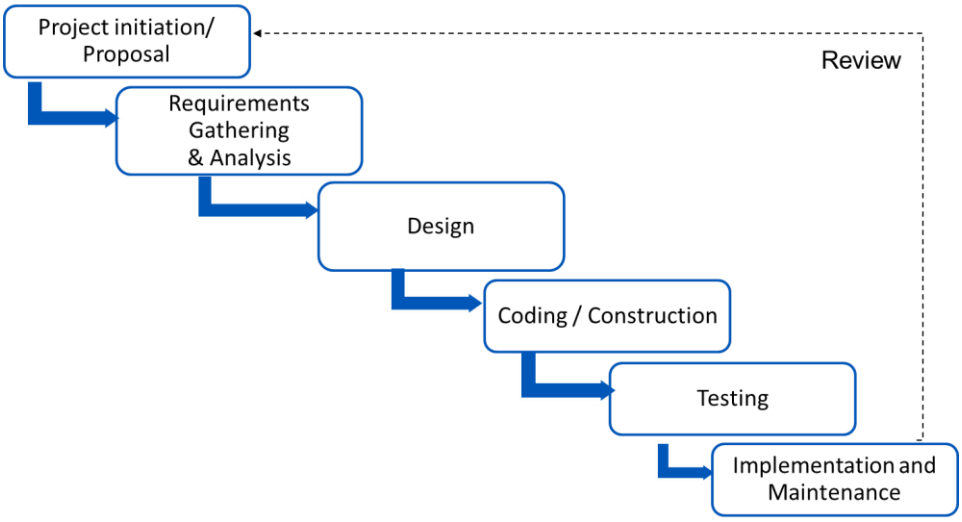
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Marks

Question 3

a) Draw and label a diagram of the Waterfall model, clearly illustrating the stages. **7**

Mark scheme



Mark Scheme
Award 1 mark for each correctly positioned stage, up to 6 marks.
Award 1 mark for appropriateness of overall diagram
Valid responses with different number of stages are acceptable.

b) Explain THREE (3) advantages of the Waterfall model. **3**

Mark scheme
Any 3 from below or any other equally valid answers:

- ***Defines end goal early. What you plan is what you get.***
- ***Provides clear structure and steps that help coordinating large projects and clarifying costs and schedules at the outset to reduce financial surprises.***
- ***Suitable for well-defined user requirements that must be defined at the outset and maintained till the end of the project.***
- ***Easier to assess fulfilment or requirements.***
- ***Insists on documentation to maintain development audit trail so development team can account for their activities.***
- ***Structured, linear approach means that the deliverables are more predictable and therefore easier to estimate and plan.***

Award 1 mark for each point up to 3 marks

Total 10 marks

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Marks

Question 4

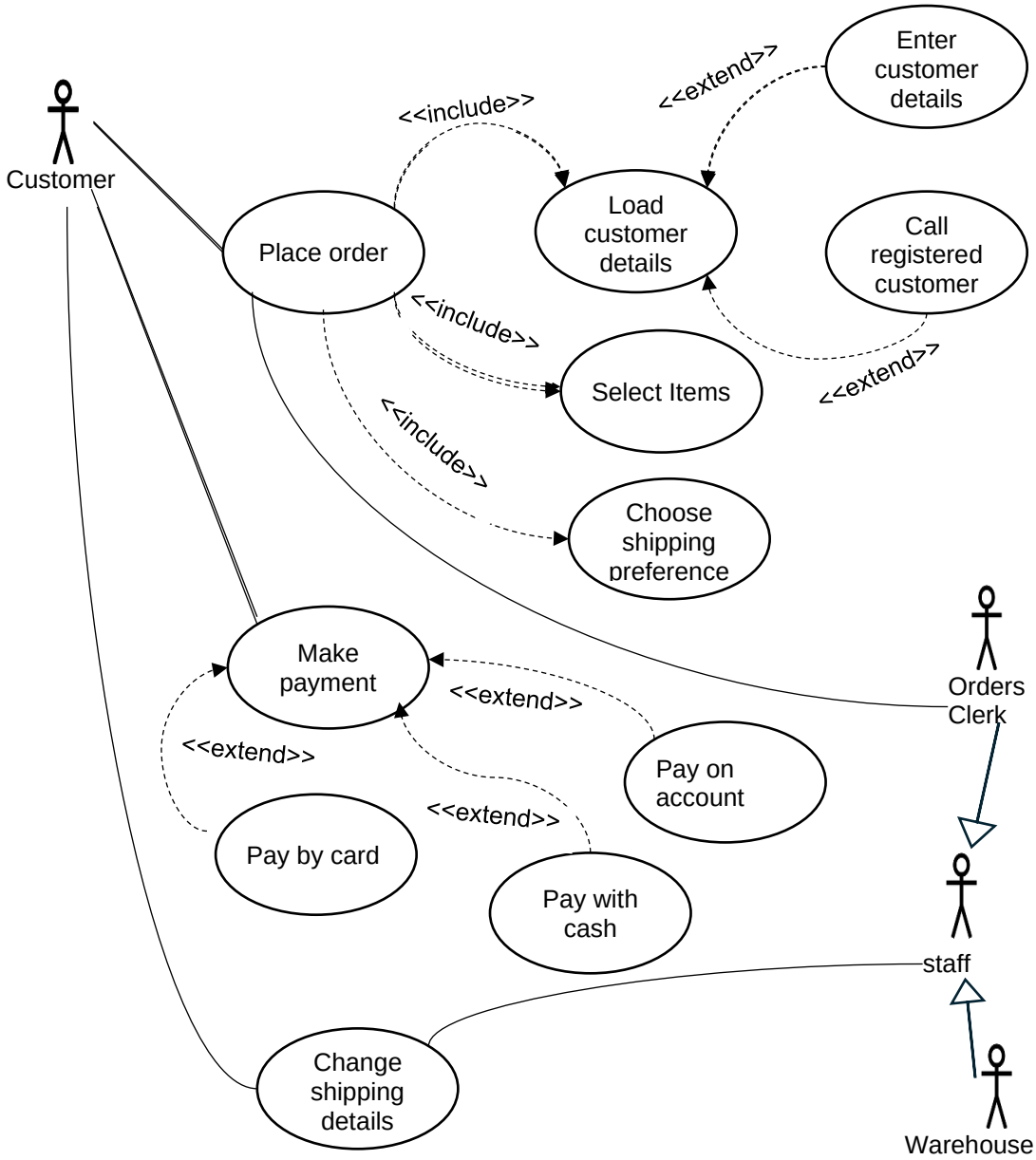
Draw a Unified Modelling Language (UML) Use Case Diagram to illustrate the activities shown in the Use Case Matrix below. When doing this, demonstrate the key concepts of generalisation, <<include>>, <<extend>>. **10**

	Customer	Orders clerk	Warehouse
Place order • Enter customer details or call up registered customer details • Select items • Choose shipping preference • Submit order	X	X	
Change shipping details	X	X	X
Make Payment • By card • Cash payment • On account	X		

Marking Scheme

- ***Award 2 marks for correct diagram syntax/ contents (use cases, actors)***
- ***Award 2 marks for consistent alignment to the matrix***
- ***Award 2 marks for generalisation. This can be on the 'Make Payment' use case or staff actor. Payment can also be used to demonstrate <<extend>> in this case. (1 mark for appropriate parent, 1 mark for children)***
- ***Award 2 marks for <<include>> on Place order as 1 mark for the appropriate major use case and 1 mark for secondary use case(s)***
- ***Award 2 marks for <<extend>> as 1 mark for the major use case, 1 mark for the secondary use cases. Major class can be 'Make Payment' or relate to Customer details. Add 1 mark if an extension point has been applied provided the overall mark for the question does not exceed 10 marks.***
- ***NB Candidates do not have to provide the complete diagram below for full marks.***

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Total 10 marks

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Marks

Question 5

a) Identify FIVE (5) metrics that can be used to assess system reliability. **5**

Mark Scheme
Any 5 from below or other equally valid items, 1 mark for each item, up to 5 marks

- ***Deviation from plan in terms of time and/or number of components deviated***
- ***Failure/error rate***
- ***Mean time to failure time (MTTF)***
- ***Mean time to repair (MTTR)***
- ***Total down time per time period***
- ***Number of defects (individual, per component /module /resource…)***
- ***Recovery time***
- ***Recovery point***

b) Identify FIVE (5) techniques that can be applied in pursuit of zero defects. **5**

Mark Scheme
Any 5 from below or other equally valid examples, 1 mark for each item, up to 5 marks

- ***Keeping it simple (KIS)***
- ***Consistency and structure***
- ***Requirements planning and continuous verification and validation***
- ***Conscientious tick-boxing as ascribed to by structured approaches***
- ***Constant communication with stakeholders as part of product development effort***
- ***Management commitment to accept cost of quality process improvement***
- ***Dynamic monitoring, logging and reporting***
- ***Resourcing within the capacity of analysis and deployment of defect prevention: staff training, workplace and time organisation.***
- ***Statistically controlled iterative approach as with the IBM Cleanroom method.***

Total 10 marks

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Marks

Question 6

Explain FIVE (5) advantages **and** FIVE (5) disadvantages of component re-use. **10**

Mark scheme
Advantages – any 5 from below or equally valid responses
- ***Time and Cost saving by reducing reinvention of the wheel and resulting in faster delivery***
- ***Product reliability especially certified/inhouse components that have been pre-tested and sometimes already trialled***
- ***Proven solutions to common problems***
- ***Opportunity to focus on user requirements not coding***
- ***Increase consistency between applications***
- ***Afford modular design which lends to easier maintenance***

Challenges – any 5 from below or equally valid responses
- ***Core requirement deficits: under or over functionality***
- ***Potential for solution bias to allow fit – may need customisation***
- ***Less control over updates, reliability, ongoing support, portability***
- ***Bugs can be replicated***
- ***If not open source, may depend on reliability of documentation***

Award up to 1 mark for each point

Total 10 marks

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Marks

Question 7

a) Define the term Testability in relation to software development. **2**

Mark scheme
Software testability is the ease with which software can be tested. (2 marks)

b) Explain FOUR (4) ways in which software testability can be assessed. **8**

Mark Scheme
Any 4 from below of equally valid responses.
- ***Operability: clear steps / operational structure***
- ***Observability: visibility of input/output including error states***
- ***Controllability: good structure and ability to manipulate states; clear boundaries***
- ***Decomposability: e.g. modularity and clear class structure***
- ***Simplicity: e.g. via cohesion, inheritance, polymorphism…***
- ***Stability: the fewer the changes, the fewer the disruptions to testing***
- ***Understandability: the more information known, the smarter the testing***

Award 1 mark for naming and 1 mark for description, up to 8 marks

Total 10 marks

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Marks

Question 8

a) Compare **and** contrast Coupling and Cohesion providing ONE (1) example of each. **6**

Mark scheme

- ***Cohesion means that a class is focussed on one well-defined purpose and methods reflect only this purpose.***
- ***Low cohesion means that a class does a number of things (1 mark)***
- ***Example: In an Estate Agency system all methods relating to appointments (make appointment, cancel appointment, reschedule) are cohesively assigned to the Appointment class and can be called by other classes. Changes to the methods are made in one place only. (2 marks)***

- ***Coupling is the degree to which a module is connected to other modules. Whilst tight coupling reflects high dependency, loose coupling means modules or components in a software system are designed to be relatively independent, with minimal reliance on each other thus enhancing reusability and maintainability. (1 mark)***
- ***Example: In an estate Agency System, methods for an Appointment are held within staff and client classes so that if there are any changes in the Appointment class, the Client and Staff classes must also be changed. (2 marks)***

b) Explain what is meant by information hiding. Provide an example to help explain your answer. **4**

Mark Scheme

- ***Implementation details of a modules (data and methods) are inaccessible to modules that have no need for such information (1 mark) so that the internal working can be changed without changing any modules that use the data (1 mark).***
- ***Other modules can use the module by simply calling it and specifying any parameters. (1 mark)***
- ***Example: A number of programs/other modules might access time by using a Clock module. Say the Clock module is called Clock() and requires a location id. Modules calling the module would simply state Clock(location id) but would have no knowledge of other variables that might be used to compute the time. (1 mark)***

Total 10 marks

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Marks

Question 9

a) Identify THREE (3) advantages of prototyping. **3**

Mark Scheme
Any three from below or any other equally valid answers:

- ***Creates models of the required system that promotes thorough verification and validation of the required system.***
- ***Lends active follow through of user needs.***
- ***Results in design, usability and maintainability improvements.***
- ***Can also result in reduced development effort.***

Award 1 mark for each point, up to 3 marks

b) Describe the evolutionary prototyping method and explain THREE (3) advantages of applying evolutionary prototyping. **7**

In your answer you should clarify how evolutionary prototyping resembles or differs from throwaway prototyping or incremental prototyping.

Mark Scheme

- ***The evolutionary prototype is actually a version of the system that is released whilst development continues through a number of iterations (1 mark). The iterations can be open-ended and might be considered to be system upgrades. (1 mark)***
- ***A working model is created at each iteration. (1 mark)***
- ***Iterations capture user requirements as they are understood and only proceeds with those requirements that are understood. (1 mark)***

<u>Any 3 advantages from below:</u>

- ***Unlike a throwaway prototype, it is re-used and adapted.***
- ***Adaptations can factor in a changing environment.***
- ***Unlike the incremental method, the whole product is developed and rolled out with each evolution being an upgrade rather than a feature increment.***
- ***Unlike a throwaway prototype, it allows phased testing in a live environment and can therefore uncover operational issues that might not be uncovered by testing of a throwaway prototype.***
- ***Like other prototyping methods, it affords user engagement throughout development.***
- ***If well managed, it can allow speedy delivery as each evolution can be rolled out for use whilst the next iteration is developed. This can further help to provide valuable feedback for the next evolution.***
- ***The points above mean that the final product is more likely to meet user requirements.***

Award 1 mark for each advantage, up to 3 points.

Total 10 marks

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Marks

Question 10

Study the following description of part of a system:

The system records information about students, tutors and units that they study in a university course.

- Students study SIX (6) units and a unit can be studied by many students.
- A unit has a code, name and duration. A unit may have many presentations, defined by a start date and the tutor who teaches it.
- Students register for a particular unit presentation. A tutor may teach many unit presentations.
- The system should record the name and university id for each student and tutor and also the course code on which the student is registered.

a) Identify FOUR (4) candidate classes from the description **4**

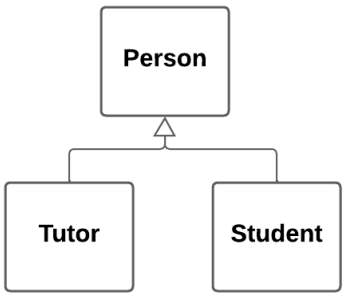
Mark Scheme

- ***Award 1 mark for each of the following to a max of 4. Names may differ for UnitPresentation and Person.***
- ***Student, Tutor, Unit, UnitPresentation, Course, Person (Abstract class)***

b) Draw a UML Class diagram showing how inheritance can be represented between classes in the system described above. You need not show attributes in your diagram. **3**

Mark Scheme

- ***1 mark for correct syntax showing inheritance symbol***
- ***1 mark for appropriate superclass (must be valid in the system above)***
- ***1 mark for appropriate sub-classes (must be valid in the system above)***

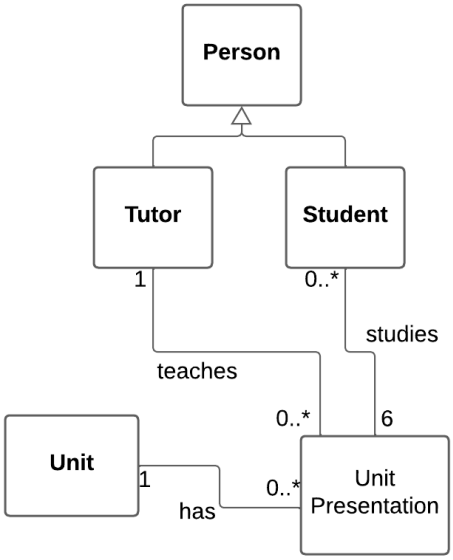


c) Draw a UML Class diagram showing an association between TWO (2) classes that you identified in Question 10a from the system described above. You should include the multiplicities in your diagram. You need not show attributes in your diagram. **3**

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Mark Scheme

- ***Marks awarded for any ONE (1) of the associations shown in the diagram below:*** `teaches, studies` ***and*** `has` ***in the diagram below. Multiplicities must be feasible. Eg * instead of 6 is ok***
- ***1 mark for correct syntax showing 2 classes related by association with multiplicities (need not be valid classes)***
- ***1 mark for appropriate classes being associated***
- ***1 mark for valid multiplicities at both ends of the association.***



Total 10 marks

END OF PAPER

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Learning Outcomes Matrix

Task	Learning Outcomes assessed	Marker can differentiate between varying levels of achievement
1	2	Yes
2	1	Yes
3	1	Yes
4	2, 3	Yes
5	4	Yes
6	1, 4	Yes
7	1, 4	Yes
8	2	Yes
9	1	Yes
10	2, 3	Yes

Grade descriptors

Learning Outcome	Fail	Referral	Pass	Merit	Distinction
1. Discuss various approaches to systems analysis and design and explain their strengths and weaknesses.	Unable to identify and assess approaches.	In an extremely limited way, can identify and assess methodologies and methods to reach limited solutions.	Can adequately identify and assess appropriate methodologies and methods to reach appropriate solutions.	Can sufficiently identify and assess appropriate methodologies and methods to reach good solutions.	Can coherently identify and assess coherent methodologies and methods to reach well-defined and structured solutions.
2. Apply object-oriented techniques for systems analysis and modelling of well-defined but complex problems.	Unable to explain key principles or discuss developments.	In an extremely limited way, can explain principles and discuss developments.	Can adequately explain principles and discuss developments.	Has very good grasp of explain principles and discuss developments.	Can coherently discuss developments and explain and adapt principles.
3. Explain and use UML to represent object-oriented models.	Little to no understanding of UML or object-oriented models.	Limited understanding of UML or object-oriented models.	Adequate ability to apply UML with sufficient representations.	Demonstrates very good ability to apply and adapt UML models with a substantial system understanding.	Outstanding ability to apply UML and adapt models in a manner that demonstrates understanding of effectiveness of methods, actions and results.
4. Understand the integration between the design models and implementation.	Lacks awareness of different perspectives or approaches within the area of study.	Has extremely limited awareness of different perspectives or approaches within the area of study.	Has adequate awareness of different perspectives or approaches within the area of study. Understands requirements traceability but with some gaps.	Has very good, informed awareness of different perspectives or approaches within the area of study. Understands requirements traceability.	Excellent awareness of different perspectives or approaches within the area of study.

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